

KOMAL SINGH

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PROFESSIONAL SUMMARY

Functional Analyst and Data Analyst with 8+ years of experience working across supply chain, retail, e-commerce, and marketing domains. Skilled at facilitating requirements workshops with business stakeholders, SI teams, and source system owners converting operational needs into clear functional specifications and validated data solutions. Proven track record of end-to-end project ownership: from requirements and data gap analysis through UAT coordination, pipeline development, and stakeholder sign-off. Hands-on with SQL, Python, cloud platforms (AWS, Snowflake, Databricks), and BI tools. Experienced supporting both BA-led system delivery and DA-led analytics programs across enterprise-scale environments.

TECHNICAL SKILLS

Languages: Python (PySpark, Pandas, NumPy), SQL, PL/SQL, Scala, R, Bash/Shell

Cloud & Warehousing: AWS (S3, Glue, Redshift, Athena, Lambda, EMR), Snowflake, GCP BigQuery, Databricks

Analytics & BI: Tableau, Power BI, Amazon QuickSight, Looker, Google Analytics, Advanced Excel

ETL/ELT & Orchestration: Airflow, Talend, Informatica, dbt, Alteryx, AWS Glue, SQL*Loader

Big Data: Apache Spark, Apache Kafka, Hive, Flink, Delta Lake, Apache Iceberg

Data Modeling: Star/Snowflake Schema, Data Vault 2.0, Medallion Architecture, Lakehouse Design

Functional Analysis: Requirements Gathering, Data Quality Validation, UAT Coordination, Functional Testing, Data Gap Analysis, SI Collaboration

Marketing Analytics: Marketing Mix Modeling (MMM), Multi-Touch Attribution (MTA), Funnel Analytics, ROI Analysis, Campaign Analytics

PROFESSIONAL EXPERIENCE

Analytics Engineer | Walmart

Jan 2024 – Present

Embedded analytics engineer and functional analyst supporting a global supply chain data platform across Finance, Operations, Supply Chain, and Marketing — acting as the bridge between business stakeholders, SI teams, and engineering.

Project 1: Supply Chain Data Platform — Requirements, UAT & Functional Delivery

- Facilitated requirements workshops with business stakeholders, product teams, and SI partners across 15+ business domains documenting functional specs, data flow requirements, and acceptance criteria that guided end-to-end platform delivery.
- Collaborated with product teams to understand business use cases and priorities, translating them into clear functional specifications that guided solution implementation and ensured delivered solutions met functional expectations.
- Partnered with source system owners to identify data gaps, quality issues, and mismatches across supply chain and retail datasets, enabling timely remediation before platform delivery.
- Validated data products against business requirements using SQL, performing data quality validation and requirement traceability checks across ingested datasets prior to stakeholder sign-off.
- Authored UAT test cases aligned to business requirements and coordinated UAT cycles with cross-functional stakeholders managing defect logs, severity triage, and go/no-go decisions across all testing phases.

Impact: Unified reporting across 15+ domains · Data availability +40% · \$5M compliance risk reduction through PII masking and lineage tracking

Project 2: Data Architecture, Dimensional Modeling & Governance

- Designed Snowflake data warehouse using Medallion Architecture (Bronze/Silver/Gold) made deliberate modeling decisions on layer boundaries, SCD handling, and grain definitions; selected star schema for Finance and Supply Chain Gold layers based on BI query concurrency patterns across Power BI and QuickSight consumers.
- Architected end-to-end AWS pipelines (S3, Glue, Redshift, EMR) with partition pruning and caching strategies, reducing query execution time by 35% and lowering EMR infrastructure costs.
- Led data governance workflow: established Glue Data Catalog ownership per domain, implemented column-level PII masking for vendor and customer data, and built end-to-end lineage tracking from source ingestion through Gold layer consumption.
- Partnered with ML engineers to design and maintain SageMaker feature stores documented feature specifications, defined freshness SLAs, and built curated pipelines ensuring training data quality, delivering an 18% improvement in model training efficiency for predictive analytics models.

Impact: Query performance –35% · Data redundancy –30% · Model training efficiency +18% · Infrastructure cost reduction through EMR optimization

Project 3: Inventory Replenishment, Order Fulfillment & Vendor Analytics

- Analyzed supply chain datasets using SQL to validate inventory, fulfillment, and vendor data against business requirements surfacing discrepancies across distribution center and retail location feeds for timely resolution.
- Developed inventory analytics pipelines tracking real-time stock levels across DCs and retail locations, enabling early identification of replenishment gaps before stockout events occurred.
- Built SQL and PySpark order fulfillment models tracking end-to-end delivery performance across the supplier-to-shelf journey identified bottlenecks in inbound freight, DC processing, and last-mile stages; presented findings with remediation recommendations to supply chain leadership.
- Built self-service dashboards in Power BI and QuickSight monitoring campaign activations, revenue generation, and performance across 15+ domains reducing time-to-insight by 40% and enabling independent root-cause analysis.
- Designed regression and forecasting models to evaluate ROI across marketing channels, supporting strategic decision-making for campaign planning and resource allocation.
- Created Power BI vendor performance dashboards monitoring lead times, delivery reliability, and fulfillment rates across 20+ suppliers enabling procurement teams to proactively escalate at-risk orders.

Impact: Stockout incidents –25% · On-time delivery rate +20% · Time-to-insight –40%

Senior Data Analyst | Amazon

Apr 2021 – Jul 2022

Senior analyst across TRMS (Transaction Risk Management) and global marketplace operations owning requirements gathering, dimensional data modeling, decision science support, operational analytics, and executive reporting.

Project 4: TRMS Seller Fraud Detection — Requirements, Modeling & Decision Science

- Collaborated with fraud investigation stakeholders to gather and document functional requirements for decision science models maintaining requirement traceability from intake through delivery and validation.
- Translated investigator heuristics into formal data model specifications and ML feature requirements; acted as liaison between business users and ML engineering, ensuring model features reflected real operational fraud patterns.
- Redesigned the performance assessment data framework for 500+ global operations using regression analysis and audit pipelines codified defect driver taxonomy across 400+ categories to support ML model training.
- Built a dimensional Redshift model for fraud analytics designed seller risk fact table with slowly changing vendor attributes, account linkage bridge table, and transaction pattern aggregates optimized for anomaly detection query patterns.
- Ran structured UAT against known-good and known-bad seller account sets; managed defect log and coordinated resolution with engineering before production deployment.
- Developed automated PySpark and SQL data validation pipelines, eliminating 30% of manual reporting overhead and improving financial data accuracy across global marketplace operations.

Impact: Seller detection rate +22% · False positive rate –28% · Investigation prep time –65% · Pipeline reliability 98.9%

Project 5: Stakeholder Communication, Executive Reporting & Ad Hoc Analytics

- Served as primary analytics contact for Operations, Finance, and Product leadership fielded and prioritized ad hoc analysis requests, translating ambiguous business questions into structured analytical problems with defined outputs and timelines.
- Delivered weekly and monthly performance reports to senior leadership synthesizing trends across fraud, fulfillment, and marketplace operations distilled complex multi-source datasets into clear narrative findings with actionable recommendations.
- Built Multi-Touch Attribution (MTA) and funnel analytics models in Python and SQL evaluating performance across marketing touchpoints surfaced channel-level ROI insights that directly informed campaign budget reallocation decisions.
- Conducted deep-dive root-cause analysis on performance anomalies across global marketplace channels structured findings into executive-ready presentations surfacing insights that drove policy and operational changes.
- Orchestrated multi-step Airflow workflows across 20+ dependent data products, reducing pipeline failure rates and enabling transparent SLA monitoring cutting ad hoc reporting requests by 35%.

Impact: Ad hoc requests –35% · Executive reporting: weekly/monthly across fraud, fulfillment, marketplace · Campaign optimization through MTA-driven channel reallocation

Data Analyst | Amazon

Apr 2019 – Mar 2021

Data analyst supporting global marketplace operations, supply chain analytics, fraud pipeline automation, and cross-functional reporting across multiple business programs.

Project 6: Supply Chain Pipeline Automation & Operational Analytics

- Partnered with SI teams and source system owners to validate data quality and identify mismatches in operational datasets supported accurate downstream analytics and requirement traceability across multiple global marketplaces.
- Facilitated requirements gathering with business and product stakeholders; supported functional testing of pipeline deliverables, validating outputs against business requirements and documenting issues for development team resolution.
- Developed Python and AWS Lambda automation pipelines streamlining 45% of manual workflows eliminated the equivalent of 6.5 FTEs in manual processing effort across operational reporting and fulfillment teams.
- Built and optimized supply chain analytics pipelines reducing data discrepancies by 20% and preventing inventory and order fulfillment delays; built SQL-based ETL processes ingesting and transforming operational data into AWS S3 and Redshift.
- Built correlation, regression, and forecasting models in Python and SQL to evaluate operational trends; integrated ML anomaly detection algorithms into existing pipelines, improving fraud prediction accuracy across global marketplaces.
- Led funnel analytics and segmentation analysis surfacing inefficiencies that drove a 25% reduction in risk exposure; collaborated with operations, finance, and product teams to define KPI frameworks tied directly to business priorities.

Impact: Manual workflows –45% (6.5 FTE equivalent) · Supply chain discrepancies –20% · Risk exposure –25%

Data Analyst | Rolon Seals

Jan 2017 – Feb 2019

Sole data and IT analyst supporting a 15-vendor industrial supply chain operation — owned the full analytics lifecycle from requirements through delivery, training, and ongoing support.

Project 7: Supply Chain Analytics Overhaul — SQL, Power BI & Report Automation

- Gathered requirements from Procurement, Operations, and Finance mapped all data sources, identified gaps, and resolved conflicting metric definitions between teams before building any solution.
- Migrated 30+ disconnected Excel files across 15+ vendor relationships into a centralized SQL database using Python (Pandas) and Power Query built a governed vendor master reference table establishing the company's first master data asset.
- Queried ERP/SAP database directly via SQL to extract live procurement and inventory data; designed and deployed 12+ Power BI dashboards covering vendor scorecards, stock level alerts, PO aging, and monthly spend replacing all legacy Excel reports.
- Conducted segmentation, correlation, and regression analysis on procurement and inventory data across 15+ vendors to surface on-time delivery patterns and reorder optimization opportunities.
- Ran UAT personally with each stakeholder sat with Procurement, Operations, and Finance individually to validate every metric against known source data before go-live; wrote user guides and trained all three teams on Power BI.
- Automated all recurring weekly and monthly reports via Python scripts; optimized SQL pipelines improving query performance by 40%, eliminating all manual reporting effort across the organization.

Impact: Manual reporting –100% (5–6 hrs/week) · Discrepancies –20% · Vendor visibility: real-time · 30+ Excel files retired · Query performance +40%

EDUCATION

Master of Science in Data Analytics | Webster University, St. Louis, MO

May 2024